

PANTH MISTRY

COMPUTER SCIENCE & ENGINEERING • ASPIRING AI DEVELOPER

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PROFESSIONAL SUMMARY

Driven Computer Science & Engineering student at ITM SLS Baroda University with rigorous foundations in algorithmic problem solving, object-oriented design, and systems development. Passionate about Artificial Intelligence, large language models, and high-performance graphics engines. Experienced in building native Android applications with Firebase, custom browser game physics engines from scratch in Vanilla JavaScript, and procedural WebGL/GLSL shader pipelines. Actively targeting a B.Tech in Artificial Intelligence to engineer intelligent, production-ready software platforms.

EDUCATION

ITM SLS Baroda University 2024 – 2027 (Expected)
Diploma in Computer Science & Engineering • 3rd Year (5th Semester) Vadodara, Gujarat, India

- **Core Coursework:** Data Structures & Algorithms (DSA), Object-Oriented Programming (OOPs in C++/Java), Database Management Systems (DBMS), Operating Systems (OS), and Artificial Intelligence Fundamentals.
- **Academic Trajectory:** Preparing for B.Tech in Artificial Intelligence with a dedication to machine learning, neural architectures, and intelligent systems.

TECHNICAL SKILLS

Programming Languages: Python, C++, JavaScript (ES6+), Java, SQL, HTML5, CSS3

AI & Data Analysis: Google Gemini API, Prompt Engineering, Large Language Models (LLMs), Pandas, Statistical Data Modeling, Advanced Microsoft Excel

Tools & Frameworks: Git, GitHub, VS Code, Android Studio, Firebase (Realtime DB & FCM), Figma (UI/UX App Design)

Core Engineering: Data Structures & Algorithms (DSA), Object-Oriented Design (OOPs), WebGL & GLSL Shaders, 2D Collision Physics, Event-Driven Architecture

FEATURED PROJECTS

Smart Vehicle Service Reminder App | Android System Java • Android XML • Firebase Realtime DB • FCM

- Engineered a proactive Android vehicle maintenance system with dynamic service interval calculations based on live odometer logs and elapsed dates.
- Integrated Firebase Cloud Messaging (FCM) for background automated push notifications and persistent maintenance reminders.
- Architected local offline storage caching with cloud synchronization for maintenance receipts and expense history.

Troll Adventure • 2D Platformer Game | Custom Physics Engine Vanilla JavaScript • HTML5 Canvas • Custom Physics

- Built a custom 2D browser platformer game from scratch without external game frameworks, implementing AABB collision detection and gravity acceleration.
- Designed enemy AI state machines with line-of-sight detection and progressive player pursuit routines.
- Optimized 60 FPS Canvas rendering using requestAnimationFrame and animated sprite sheets.

Castle Warrior • Action Wave Defense | College Game Jam JavaScript • HTML5 Canvas • Wave AI Logic

- Developed an action-packed defense combat game within competition time limits during the university Game Jam.
- Implemented procedural wave spawn algorithms with escalating difficulty curves, melee hitboxes, and inventory mechanics.

Ferrofluid Mathematical Shader Simulation | Creative Tech WebGL • GLSL • TypeScript • Value & Derivative Noise

- Authored hardware-accelerated GLSL fragment shaders simulating magnetic fluid turbulence and smooth-minimum wavefield blending.
- Maintained 60 FPS hardware acceleration with dynamic pointer interactions and specular rim lighting.

CERTIFICATIONS

- **Data Analytics Certificate** – Statistical modeling, variance analysis & regression (2026)
- **LLM Learning Certificate** – Google Gemini Labs prompt engineering & reasoning (2025/2026)
- **Computer Coding Course** – Comprehensive programming foundations & algorithms (2024)
- **Designathon Recognition** – UI/UX prototype evaluation (2026)

WORKSHOPS & HONORS

- **Google Gemini Labs** – Hands-on prompt workflows, chain-of-thought & agents
- **University Game Jam 2026** – Built rapid game prototypes under strict deadlines
- **College Game Jam 2025** – Developed Castle Warrior combat engine

- **Figma App Designing Competition** – Mobile UI/UX architecture & interaction

ADDITIONAL INFORMATION

Professional Strengths: Analytical Problem Solving, Fast Learner, Systems Thinking, Team Collaboration, Disciplined Execution.

Languages: English (Professional Working), Hindi (Fluent), Gujarati (Native).